Detection and Analysis of Exoplanets Using Stellar Brightness Data

Abstract

This report studies brightness observations of STAC-1, STAC-2 and STAC-3 to identify possible exoplanets. We estimated transit depth and orbital period from the dips, then calculated planetary radius and orbital distance. STAC-3 was selected for further investigation, although its observations are not fully regular.

Index Terms— Exoplanets, transit method, stellar brightness, orbital period, planetary radius, habitable zone.

I. INTRODUCTION

An exoplanet is a planet orbiting a star outside our Solar System. A planet can block a small amount of light when it passes in front of its star. This is called a transit. Repeated dips may suggest the presence of a planet.

II. METHOD

First, we plotted relative brightness against time for each star. We looked for repeating dips and calculated their spacing. The equations used were:

Transit depth = (normal brightness - dip brightness) / normal brightness.

Planet radius = star radius × square root of transit depth.

Orbital distance a = $(M \times P^2)^{(1/3)}$, with P in years, M in solar masses and a in AU.

Habitable-zone reference distance = square root of (L / solar luminosity) AU.

We used solar radius = 696,000 km, Earth radius = 6,371 km, 1 AU = 149.6 million km and 1 year = 365.25 days.

III. TRANSIT OBSERVATIONS

A. STAC-1

Dips are visible on days 3, 8, 12, 16 and 20. Most dips are four days apart, so the estimated orbital period is about 4 days. The transit depth is 1 - 0.9970 = 0.003.

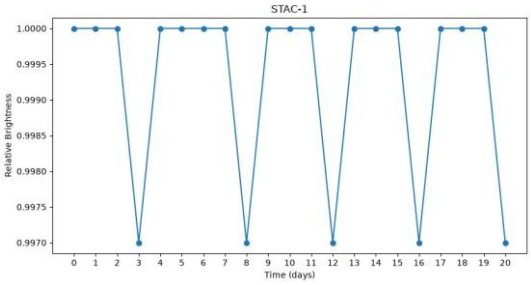


Fig. 1. Brightness of STAC-1 over 20 days.

B. STAC-2

Dips occur on days 20, 100 and 180. The gaps are 80 days, so the estimated period is 80 days. The transit depth is 1 - 0.9990 = 0.001.

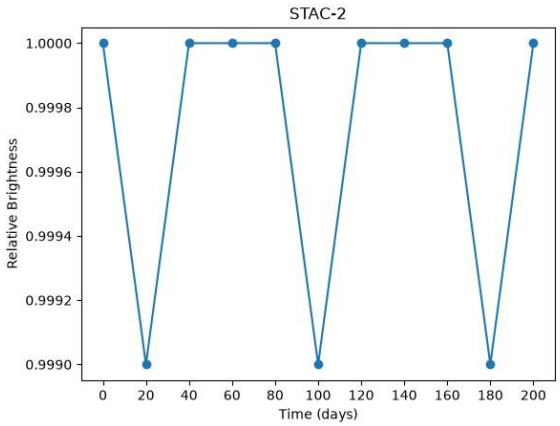


Fig. 2. Brightness of STAC-2 over 200 days.

C. STAC-3

Dips occur on days 20, 60, 100, 140 and 170. The first four indicate a 40-day period, but the final dip is only 30 days after the previous one. Average depth = (0.0010 + 0.0008 + 0.0009 + 0.0007 + 0.0010) / 5 = 0.00088.

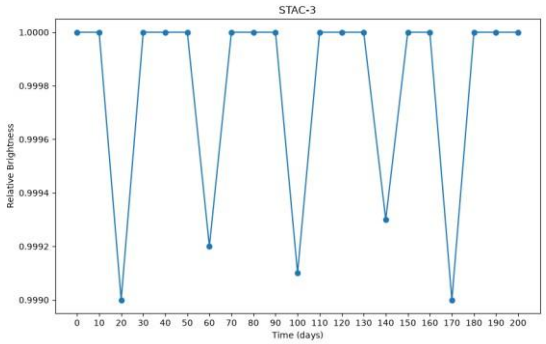


Fig. 3. Brightness of STAC-3 over 200 days.

IV. CALCULATIONS

STAC-1: Radius = 696,000 × sqrt(0.003) = 38,121.49 km = 5.98 Earth radii. Period = 4/365.25 years. Orbital distance = $(1.00 \times (4/365.25)^2)^{(1/3)}$ = 0.0493 AU = 7.38 million km.

STAC-2: Radius = (0.80 × 696,000) × sqrt(0.001) = 17,607.56 km = 2.76 Earth radii. Orbital distance = $(0.80 \times (80/365.25)^2)^{(1/3)}$ = 0.3373 AU = 50.46 million km.

STAC-3: Radius = (0.50 × 696,000) × sqrt(0.00088) = 10,323.35 km = 1.62 Earth radii. Orbital distance = $(0.50 \times (40/365.25)^2)^{(1/3)}$ = 0.1817 AU = 27.18 million km.

V. DISCUSSION

STAC-1 and STAC-2 are approximately Neptune-sized using the size ranges chosen for this assignment, while STAC-3 is a SuperEarth. The simplified habitable-zone reference distances are 1.000, 0.632 and 0.283 AU for STAC-1, STAC-2 and STAC-3 respectively.

STAC-3 seems most suitable for further study because (1) its radius is the smallest and closer to Earth's, (2) its orbital distance is closest to the reference distance relative to the other candidates, and (3) its star has low luminosity, making its reference distance closer to the star. The last two points are connected (4) STAC-3 receives approximately 2.42 times the stellar energy received by Earth, which is the lowest among the three candidates. This makes it relatively more promising for further habitability studies, since receiving excessive stellar energy can lead to very high temperatures. but the energy received is still greater than Earth's, and further observations are required to determine whether liquid water could exist on the planet. This does not establish habitability. More frequent brightness observations are needed to check the irregular transit timing. Transmission spectroscopy could later help study its atmosphere.

Another important point is that the observations of STAC-3 are not completely regular. The brightness dips occur at days 20, 60, 100, 140 and 170. The first four observations suggest an orbital period of 40 days, but the last dip occurs after only 30 days. This could be due to measurement errors, changes in the star's brightness or another possible signal. Therefore, more observations are needed to confirm the orbital period.

Also, although STAC-3 is classified as a Super-Earth based on its radius, we cannot say for sure that it has a rocky surface.

Another limitation is that the actual surface temperature cannot be calculated from the data we have. Factors such as the planet's atmosphere, reflectivity and greenhouse effect also affect its temperature. Therefore, even though STAC-3 appears to be the most promising candidate among the three, we cannot confirm that it supports liquid water or life.

VI. CONCLUSION

All three stars show possible transit dips. Their estimated periods are approximately 4, 80 and 40 days. STAC-3 is the most interesting candidate for follow-up, but the available data are not enough to confirm habitability.

APPENDIX: SUMMARY OF RESULTS

Parameter	STAC-1	STAC-2	STAC-3
Transit depth	0.003000	0.001000	0.000880
Planet radius (km)	38,121.49 km	17,607.56 km	10,323.35 km
Planet radius (Earth radii)	5.98 $R_\oplus$	2.76 $R_\oplus$	1.62 $R_\oplus$
Classification	Neptune-sized	Neptune-sized	Super-Earth
Orbital distance (AU)	0.049 AU	0.337 AU	0.182 AU
Orbital distance (km)	7,377,537.709 km	50,461,648.114 km	27,179,081.303 km
Relative to Earth	20.278× closer	2.965× closer	5.504× closer
Habitable zone reference distance	1.000 AU	0.632 AU	0.283 AU
H/Z ratio (orbital distance / HZ)	0.05	0.53	0.64
Stellar energy received (Earth = 1)	416.49	3.52	2.42